

Multi-view Interaction Reliability Filter (MIRF) Architecture

Axiomatic Synthesis of Statistical Prediction, Semantic Auditing, and Temporal Burst Signals

Branch 1: Statistical Consistency (RRFN)

- Anchor Subsets: $\mathcal{A}_1, \dots, \mathcal{A}_5 \subset \mathcal{I}$
- Stochastic Transition Matrix: $T \in \mathbb{R}^{5 \times 5}$
- Tikhonov Pseudo-Inverse: $T_\lambda^\dagger = (T^T T + \lambda I)^{-1} T^T$
- Prediction Divergence: $|\tilde{r}_{ui} - \mathbb{E}_{\mathbf{p}}[Y^*]|$

$$\mathbf{R}_{\text{RRFN}}(\mathbf{u}, \mathbf{i}) = \exp\left(-\frac{|\tilde{r}_{ui} - \mathbb{E}[Y^*]|^2}{2\sigma_{\text{RRFN}}^2}\right)$$

Branch 2: Semantic Review Audit (LLM)

- Review Text Ingestion: $S_{ui} \in \mathcal{S}$
- Prompt Sentiment & Plausibility Verification
- Repetition & Shilling Bot Signature Detection
- Thread-Safe SQLite WAL Persistent Caching

$$\mathbf{R}_{\text{LLM}}(\mathbf{u}, \mathbf{i}) = \text{LLM_Audit}(\mathbf{S}_{ui}, \tilde{r}_{ui}) \in [0, 1]$$

Branch 3: Burst Detection (2D BLAS)

- Rolling Window Acceleration: $Z_{\text{temp}}(i, t)$
- Extreme Rating Skew: $S_{\text{pol}} = \frac{|N_1 - N_5|}{N_1 + N_5 + \epsilon}$
- Vectorized 2D NumPy Matrix Multiplication
- Compute Time: < 25ms over 42K interactions

$$\mathbf{R}_{\text{bomb}}(\mathbf{u}, \mathbf{i}) = 1 - \sigma(\omega_1 Z_{\text{temp}} + \omega_2 S_{\text{pol}} - \theta)$$

MIRF Convex Fusion & Reliability Weight Synthesis

$$\mathbf{w}_{ui} = \mathbf{R}_{\text{MIRF}}(\mathbf{u}, \mathbf{i}) = \alpha \mathbf{R}_{\text{RRFN}}(\mathbf{u}, \mathbf{i}) + \beta \mathbf{R}_{\text{LLM}}(\mathbf{u}, \mathbf{i}) + \gamma \mathbf{R}_{\text{bomb}}(\mathbf{u}, \mathbf{i}), \quad \alpha + \beta + \gamma = 1$$

Default Defensive Calibration: $\alpha = 0.40, \beta = 0.30, \gamma = 0.30 \rightarrow$ Continuous Sample Reliability $w_{ui} \in [0, 1]$

Consumer 1: Graph Contrastive Learning

Modulates InfoNCE Loss $\mathcal{L}_{\text{cl}} = -\sum w_i \log(\cdot)$ and Risk Loss $\mathcal{L}_{\text{risk}}$

Consumer 2: SUPER Blueprint Denoising

Filters Poisoned Edges: $B_u = \{i \mid w_{ui} \geq \theta_{\text{trust}}\} \rightarrow$ Pure Tail Inclination α_u